SECOND DEFENSE REPORT

Lossless inc.

Eden Lumbroso, Amélie Nannette, Pierre Dubrulle, Tarek Zampalegre, Lucas Boidin - A3

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I) The Story

During the first meeting of our group, we decided the story we wanted to represent in our game. We wanted to study the path of grief, for a little girl growing up experiencing that feeling. We find ourselves brought into the life of a small family, a father and a daughter, living in a small house in the forest. After rushing to see her dad, the young girl, Belladona, bumps into a candle, which becomes the turning point of the story. The entire house burns very rapidly, trapping inside the dad, who managed to get his daughter out of the place in a short time. Ordered by her father, she ran through the forest to find shelter, but encountered a new character, a lost heart. Filled with sadness, fear, and remorse of leaving her father, Belladona decides that this heart will be her companion during her travels. After some time, both of them decided to stop at a river to rest, but as they were bending over to reach the water, Belladona fell into it. Controlled by the current, she couldn't stop and realized that she was going to plunge into a waterfall. Without managing to carry herself out of the water, she began her downfall, passing out during her drop. When she finally wakes up, she is lying on the ground, soaked in water, the heart sitting next to her. From that point, Belladona must make her own path to reach the end of her journey, hopefully a better place. She walks through seven worlds, the seven stages of grief as we know. For each of these worlds, she is going to have to fight her own enemies: the villains that only want to retain her from growing, evolving, and moving on. Both characters follow the river up and will be confronted to many tedious and bizarre adventures.

II) Eden’s part

A) Graphics

1) Backgrounds

Since the last defense, I have realized the background of the next world. This world focuses on the feeling of bargaining, so a place of mind where our character will try to make sense of her grief with only “what if”s and “if only”s. This includes 4 different rooms, with different layers each time. We have 3-layer types for this world, each distinct from one another. In order to represent the perspective, we want to show here, I did different color shades for each layer, from a very strong purple to a very light one. In total, I have used 15 layers, mixing the colors, but also the constructions. The rooms of this world are inside of a castle. On some layers we can see interior furniture like a sofa and bookshelves, on others we see a balcony, a special light design. Outside of the windows the view of the first forest can be seen in the beginning of the game, in the original menu. This background reminds Belladonna of her home with her dad, so an important feeling in her path through grief. This is a place where she used to find comfort and understanding, but now it’s only a place of mismatch and sadness.

2) Sprites

I also realized a couple of sprites for the enemies. I have done one enemy of the bargaining world, a bat wearing a jester hat. This enemy had a fly mode, an attack mode, and a death one. The fly one is just a normal flying bat, but the attack is a bit more special. It retracts to prepare its charge and then strikes with its back feet on the character. For the death, this enemy stops flying and drops to the ground, just like a dying bird. This jester bat is one of the many small enemies that Belladonna is going to have to fight in order to grow and move on.

I have also drawn the sprite of a nutria, the final boss of the denial world. This enemy has one more attack mode than the previous enemy. It has the normal walk mode, the attack with its clawed paw, the dash attack and death. For the normal walk, we decided to create a golden armor that can be found on every small enemy in the denial world. This is a reminder of every boss Belladonna who is going to have to fight. This shows repetition and tirelessness from the enemies but also from our characters, as they will be obliged to fight them to evolve. The normal attack is a basic paw hit; the beast retracts its paw only to charge stronger. The dash is another attack mode, the nutria charges fastly towards Belladonna, a hit that usually cannot

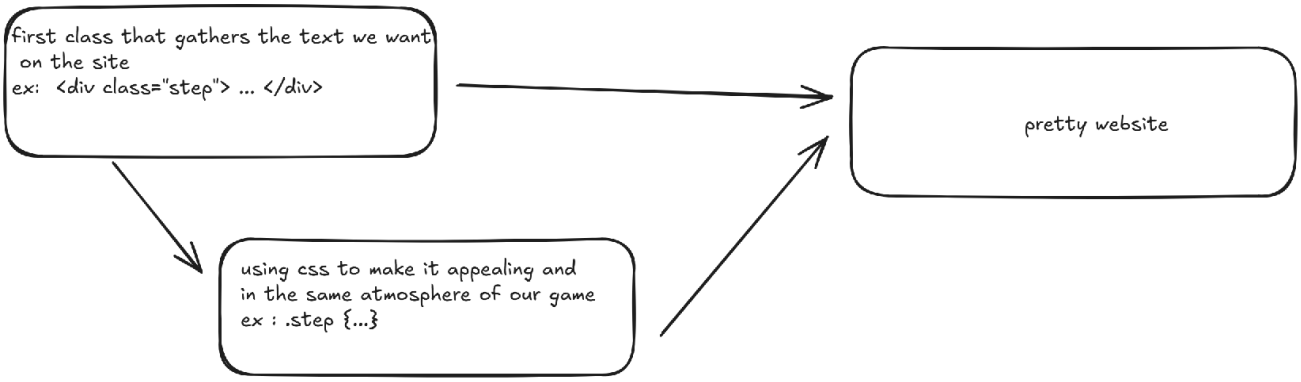
be ignored or avoided. The death sprite is the animal falling on the ground, and the hit forces it to break its long teethes on the ground, showing the end of a chapter for Belladonna. These teethes were used as a weapon, and the fact that they broke means the evolution and growth of our protagonist.

B) Website

I have coded the entirety of the website in both html and css.

1) HTML

HTML is a language that was used here only as a structure tool to gather all the information we wanted to show on our website. It joined the text and images within different classes, only forming the base of the site. As mentioned, I used classes so that the CSS part is easier to determine.

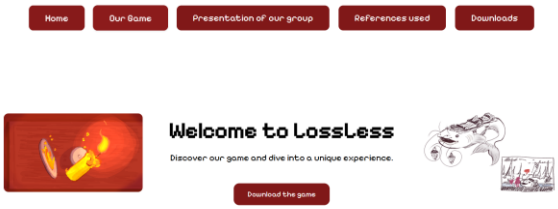


On the website, I have included top buttons to reach subpages for Our Game, the presentation of Our Group, the References used (the books and help we got), and a page to Download our previous defenses. For all of these pages I have included a footer identical for every of these pages to contact us, on our Instagram page, our github repository, and our email in case anyone wants to communicate with the team. On the main page, we have a welcoming word, but also a button to download the game when it is finished. This button will also be put in the Download page, with the defenses. When scrolling on the home page, we see 3 categories that describe the game, the 3 main characteristics of our project, with a couple of images that remind us of it. Then the bottom of the main page is a story line on a scheme that shows the different stages of grief we represented in the game. On the “Our Game” page, we are representing our game with pictures of it and different descriptions for each world. The images and text alternate, to show the different steps of the game. I have included one shot of the game for the first three worlds, as the 2 last are not yet done. Just to have the structure done and no error, I included the pictures of the lobby, but these images are not permanent. To recap this page, we will have an image for each world and a paragraph explaining our motivations and description of the game. On the next page, we have the presentation of our

group. I added a small description of each of us, our roles and main domains we are comfortable in, as well as our pictures. This page is not finished yet; I will develop the CSS more for the next defense. So, on this page we will have our personalized picture, our strong points, and a paragraph to explain our group. On the following page I created a space that gathers all of the references used to develop the game. This page will include the books we found useful, but also the videos that helped us understand Pygame and all of the elements that contributed to the elaboration of our project. On the last page, I uploaded a zone to download the previous reports of defenses and the game, as mentioned earlier. I will also add some pictures of the game and descriptions of each document. To recap the entire website, we have 5 pages, each serving its own purpose, with on each subpage some pictures and details of what is on.

2) CSS

The CSS language is used to make the website appealing, and to go further than only information communication. This language allowed me to create the style for the buttons for example. Each button has its own rectangular red zone, and when putting the mouse on each of the buttons, it appears more clearly. Each button has rounded corners, as well as the pictures in the entire website.



The CSS also made it possible to have a “post it” style for the 3 categories when scrolling. The language also allowed me to center everything, like the texts in each button, but also the footer, as well as the images, ... On the page for the description of our game, the alternance of images and text was also made thanks to the css. On a more technical aspect, the classes created on the html part were reused on the css side, with each one having different properties to determine the way I wanted to create things.

```
.page-title {
    text-align: center;
    font-size: 3rem;
    margin: 50px 0;
    font-family: "Pixelify Sans", sans-serif;
}
```

For example, here we can see the class. page-title that was originally created in the html code, but precised in CSS. These properties are used to do: it aligns the title in the center of the page, it determines the font size (based on the root size 1rem = 16px, depends on the size of the page), it also creates the size of the margins (50 px top and bottom, 0 left and right) and lastly precises the font used for this title.

3) Testing

Before reaching this final look for the home page, I did multiple tryouts, and asked for the teams’ opinion, and decided to go for this version. The “Our Group” page is also commonly decided, but I need to do the same for the last pages, so that the website is suitable for everyone.

C) Scripts

1) Long introduction

As mentioned in the previous defense, the scripts are an important aspect of the game, as they precise the story behind the images. Our first introduction to the flashbacks is a direct conversation between Belladonna and the Heart. The heart asks Belladonna how she managed to arrive at the bottom of a waterfall, and so the flashbacks will begin. So, at first, a small talk takes place, with the heart asking, “So what got you here anyways?”. Belladonna answers: “ I don't really know everything is Kind of Blurry Now”, but the Heart insists with “Take it slow and try to remember”. From that the flashbacks begin. As the flashbacks scroll, Belladonna makes comments and tells her story to the Heart at the same time. She starts with “ I remember going fishing on the lake with my dad, we used to do that all the time…”, to continue with “We got home after that and we were really tired, and I remember being very excited to talk to my dad about something. I bumped into a candle on my way there.” In the next flashbacks, Belladonna tries to remember: “From then on, all I could see were flames. The fire spread so quickly, we couldn’t do anything”. This was followed by “Everything burned. Dad managed to get me out, but he couldn’t make it.”. After some time, she finally finds her words again:” I think my father ordered me to run away far from the flames to find shelter, and so I did. I ran until reaching a river. Then:” After that, it was all fuzzy. I remember the cold feeling of the water, the sound of a waterfall, and me struggling to get back on land, then, nothing.” Right after that, a flash occurs, leading to the conversation on the Merchants Creek. The Heart explains it in the following way: “So that’s how you got here? I’m just glad I found you here... Let’s try to get back there! Maybe your dad made it? I’ll come with you. As you can see around you, this creek is quite a busy place. Are these animals around merchants, I think? They sold me these weird boxes with cards inside of them. Here you can take one!” and Belladonna then answers: “Right thanks, I guess we are in the Merchants Creek then, let’s have a look around”

The game begins from that moment; the characters can be played by both players.

D) Music Ending music

The music for the generic is also being made by me on the piano, and I have decided to play the 3d impromptu of Schubert. I am still practicing, so I will upload it for the third defense

E) What needs to be done?

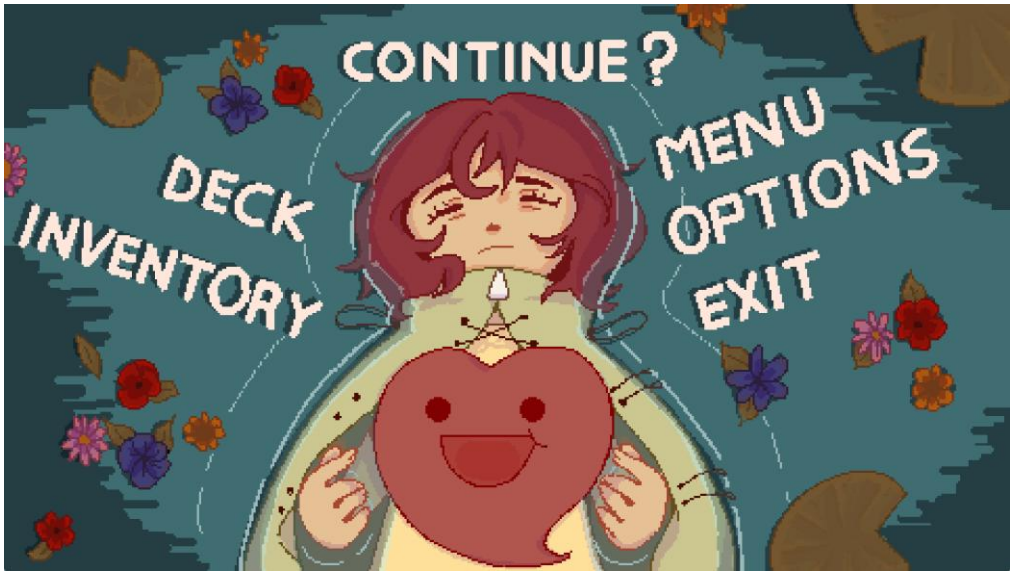
For the next defense, the game will be fully done. On my side, the website will be completely finished, so all the subpages will be filled with all of the information we want to put in. The ending generic will be done by me, so our name and acknowledgments will be scrolling down. The ending music will be playing at the same time. For another coding aspect, I will code the text implementation for the flashbacks and every part of the text appearance in the game.

III) Amélie’s part

A) The UI of the game

One of my main tasks in this project is handling the displaying, therefore the UI of our video games. This time I focused on the pause menu during the gameplay as well as some quality-of-life adjustments of the general UI.

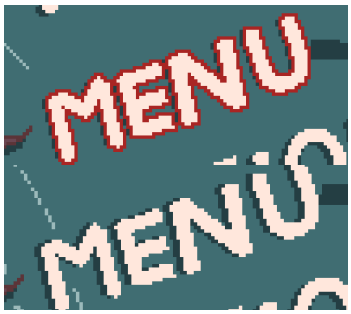
1) Pause Menu



Animation in pygame is quite complex since there is no built-in program or tool that helps with this, instead I created (at the start of the year) an animation class that takes a list of images, in our case the images represent the different frames of the animation. And with every game loop the frame of the animation is calculated and taken from the list of images, and the image at that frame is displayed on the screen. This allowed me to make different animated backgrounds in our video game

The pause menu is supposed to be the screen where information and the navigation between the gameplay and the rest of the interface is the smoothest. Making something visually appealing as well as practical was therefore necessary for us, since we first wanted to show the spirit of the game in every corner. The pause menu was actually one of the first pages I conceptually designed. I got inspired by the famous painting Ophelia by Millais, as the menu depicts the two main characters floating down in the river. This river is filled with vegetation such as flowers and lily pads. This is one of the only depictions of Belladonna with her jacket open, revealing the yellow shirt she wore in the introduction sequence. This makes her more vulnerable, reminiscing of the scene of Ophelia in the painting or in the story as if she was drifting to her death. Her expression reflects the melancholy she feels after the tragedy that happened to her. The Heart (our player 2) is smiling blissfully in the middle, over the heart of Belladonna contrasts with her own expression.

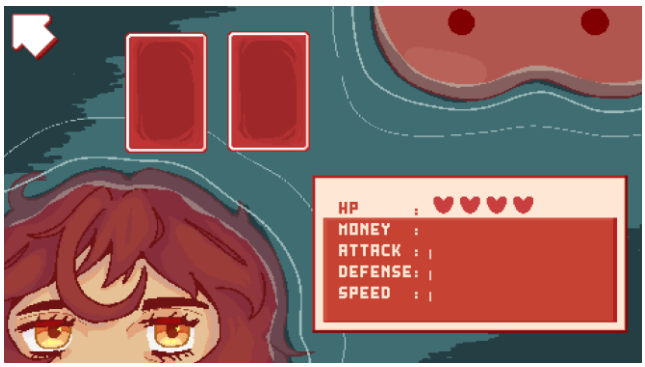
a) Buttons

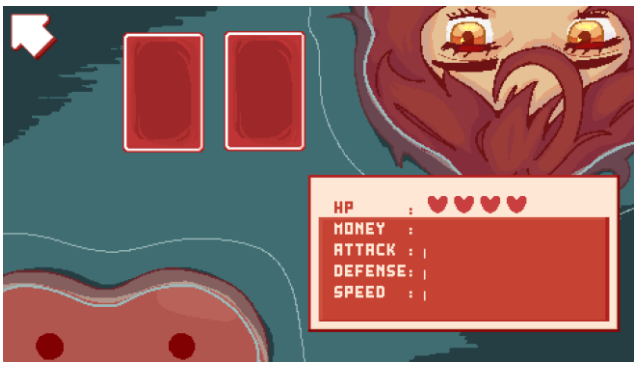


Buttons in pygame, in our game, are made by creating an object, a button which initializes the coordinates of the button as well as its image. Here we also have a hovering image, which is a visual cue, an image that indicates when the mouse of the player is on top of the button. Here the button just always checks for collision with the mouse of the player and if we click on it, something, which depends on the button, happens.

The buttons are disposed around the character as if they were floating in the surrounding water. This was made to give a better depth to the water. At the top of the screen, we have the continue button which brings the player back to the game, on the right we have the menu button which brings the player back to the main menu, we also have the options page as well as exit. Thus, on the left we have the more technical part of the menu being about navigating and modifying the experience of the game. On the right part we have the more gameplay part of the game, with the buttons for the inventory and the deck.

b) Inventory





Player 1 **Player 2**

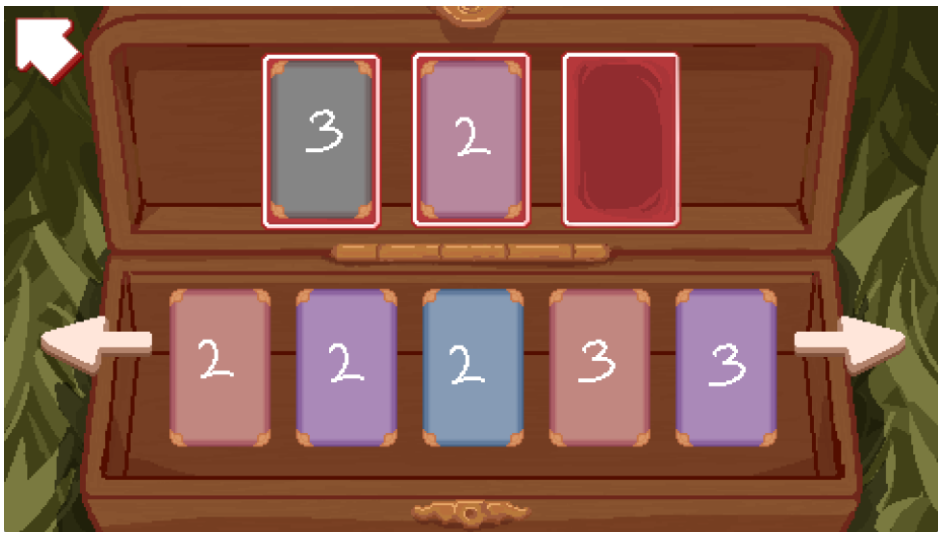
The inventory page was quite difficult to make conceptually, since we still only have an idea of what we want it to do, which is show the player the current item they possess, as well as their stats. At first, I settled for a dark overlay (as you can see on the left) over the gameplay. The inventory would show the weapon, and the stats of the player in a black box. This design did not feel right with me for a while, but I only recently changed it since I realized it did not fit at all with the vision and the overall idea we had of the game. So, I came

back to the symbolism of the river. This time I chose to focus on what we already had which is why I took inspiration from what I had done on the join and play page of the game, but this time the background would display both characters at each end of the screen. This was made so that if we were playing as player one, we could have her on the bottom left on the screen and the heart on the top right. And when we play as player two, the background would flip, therefore the characters would flip as well.

For the information displayed in the inventory, we have the stats, which are the attack, defense and speed which can be added on or changed with cards (cf. A) 1). c).). We also have the money which will be shared between the two players. As well as the current health of the player. In the inventory we also have two slots, one for the currently equipped weapon and another one for the currently equipped card.

For now, the inventory is purely visual as it doesn't show any of the currently selected and held items, which is something that I will tackle soon.

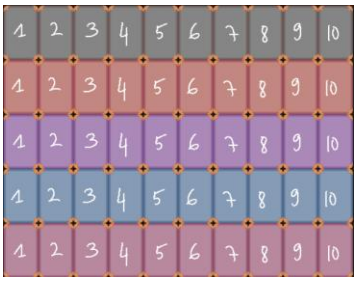
c) Deck and Cards



In our game, the cards represent different bonuses the player can equip, mostly it will give the player stat boosts or advantages, and the way the cards chosen in every game will be decided is by luck at the start of each game of a world. In the deck the player will choose 3 cards from the ones they have, these three cards will have a chance to be selected at the start of each game. Cards are dropped at the end of every boss fight only for player 1. Player 2 can get cards only if player 1 sacrifices one of their cards to a non-playable character named the oyster lady. Therefore, the deck is the place where their cards are stocked. Every player has their own deck with their own cards.

For the implementation of these cards, there are a sprite sheet of 10 cards by 5 cards (as we can see on the left which is a placeholder), each player has an object deck of their own which is a list of tuples stocking the stat increase or the effect of the card as well as the card itself. We can therefore add and remove cards from this list. The cards possessed by the player are displayed at the bottom of the screen and the player can choose cards among this

selection which will be displayed on the top of the screen. So far, the cards are implemented and so are their stats, I just have yet to make the final design as well as have the cards have an effect on the player and be chosen randomly at the start of each game. For which I will use the random library in python.



For the design of the deck, I had the idea quite early on what I wanted to do for it. I knew it had to be a wooden box with cards, which is exactly what I ended up doing, the box is lying on grass. The players will get this box in the opening cutscene of the game where the Heart (player 2) will expose the context of the game to the girl (player 1).

2) Popups

Popups are important for quick information for the player so they know what is happening, it can be easily overlooked as something simple but is actually quite important to the design and navigation of a game or a system in general.

I designed popups for when the connection or the launching of the server fails, the player will therefore get informed about what is wrong with the game instead of just wondering why it could not launch. This could be shown if when launching the server, the player is not connected to the Wi-Fi or the player inputted the wrong IP address.

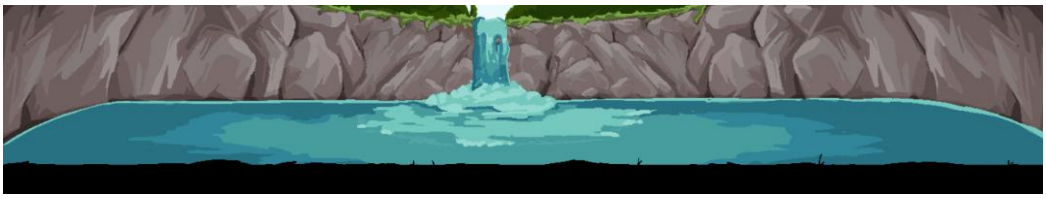
I also designed what I call “are you sure” popups which confirms the player's action in the game, it is mostly there for exits from the pause menu to either go to the menu or quit the game. It will confirm the player wanted to leave since this is a two-player game, leaving in the middle of a game can be inconvenient, therefore double checking.

Technically, the popups freeze the game since it creates a new loop displaying only the popup over the screen, where we only verify the player input to either leave or remove the popup which forces a response from the player.

3) Merchant’s Creek

Merchant’s Creek is the hub of the game, of the gameplay, this is where the players are free from attacks and enemies and can regroup, strategize and move across the map or trade using the many

non-playable characters (NPC) scattered across. We settled for 3 NPC, one being the oyster lady which can create cards for the player 2 from the cards of the player 1 (cf. A) 1). c).), the weapon merchant who will hold the non-used weapons of the players and the travelling fish who will be able to drop off the players to every world they have already visited. I designed the sprite sheet of the weapon merchant and the travelling fish so far (cf. B) 3). and 4).) and I plan on finishing those of the oyster lady and the travelling merchant soon.



Eden and I both worked on designing Merchant's Creek. Since we wanted something to do with water, fish and of course the river, we settled on a giant lake at the bottom of a waterfall. The area is separated into three parts: the leftmost area, the middle with the waterfall and the right most area which is the entrance to the Denial world. At first, we wanted this area to be a giant continuing image where the dock would scroll in the background, but due to technical restrictions from the way the rooms are implemented we settled for three different areas with transitions.

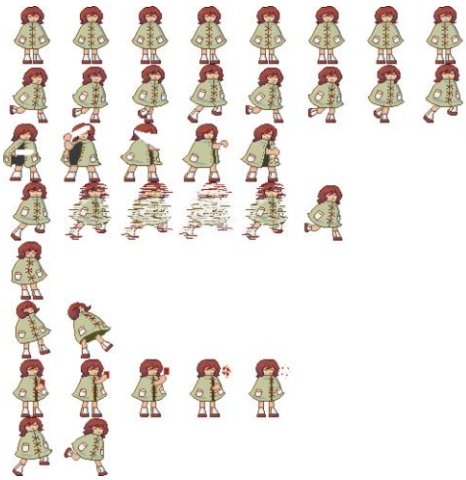
I designed the rocks as well as the animation of the water in the background, which I created as a whole piece at first and then cut into parts. I used the code Pierre made for the predefined rooms to implement the backgrounds into the game.

B) Sprites

As the lead graphic designer, I had a hand in designing sprites for the game, sprites are important since we really wanted to have a variety of enemies and interesting characters.

1) Belladonna

I continued designing the player 1 character: Belladonna. At first, we made only their standing and walking animation which was enough for what we wanted, but for the final game we wanted a lot more animation such as an attacking animation, a sprinting animation, a taking damage animation, a dead animation, a card animation and finally a jumping animation.



Initially, I did not want to have her open her jacket since it represents her being closed off to the world and because I did not want to animate arms, but to have her attack or take out a card it was necessary to show an arm movement at least for what we wanted. As such I animated her open her coat for the attack animation, we also have her arm sticking out of her coat for when she is using a card.

2) Anger

I was tasked with the making of all the sprites in the Anger world, in each world we have three enemies, one normal one and one strong enemy which are the ones you find in the rooms, the strong enemy is slightly more powerful and rarer and difficult than the normal enemy and drops better items when killed. At the end of each world, we have a boss that will test the players skills and will drop a card for the player 1 (cf. A). 1). c).) , as there are multiple cards to obtain the players are incentivized to replay the rooms multiple times.

For the story, the Anger world used to be a sprawling city that was in the middle of its construction, most inhabitants were thus construction workers. When a giant beast appeared they all got turned into black goop monsters, leaving behind a destroyed city of black goop people, while the giant beast was eating the city, ravaging it slowly over time.

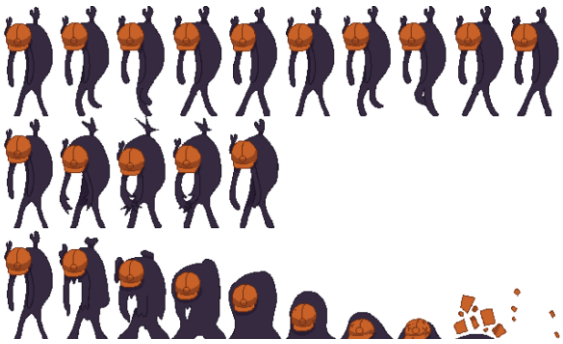
The giant beast of anger is never fought directly, it's a giant looming beast, constantly in the background and is a literal representation of anger, and how it slowly destroys one's inside mind and feeds from it getting bigger and more destructive.

a) Normal ennemi

These were the first monsters I made for the anger world, so I had a lot of different ideas. I knew I wanted them to be black goop people, since it would be easy to make a mirror as well to the current condition of Belladonna's mind. I went through multiple iterations of this monster, first it had no legs and was a big blob of black goop, but it looked out of place therefore I settled on giving it limbs. Since these were construction workers, I needed to find a way to make them look like one without too much detail which is why I settled on the construction hat which could be a characteristic of the monsters of this world.

The death part of this monster was actually the easiest to make and think of, since it was the monster fading into a puddle, returning to the giant beast It came from. The only hard part was to figure out what to do with the hat when it dissolved, so I made it explode into smithereens.

b) Strong ennemi

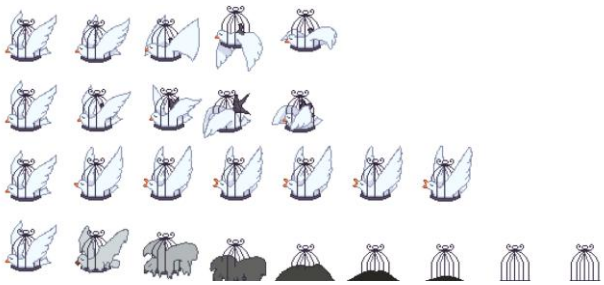


This one was quite harder to create than the first one as it had to fit the theme and idea while looking stronger and tougher. Since they came from the same idea and theme, I still made a big blob of black goop with a construction hat and deer horns but it was taller and gloomier.

One narrative idea I wanted to portray was that the goop was violent and had a mind of its own. Thus, I added a spike on the back of this monster that straightened whenever it attacked to show violence. Finally, the idea for the death was the same as the first one.

c) Boss

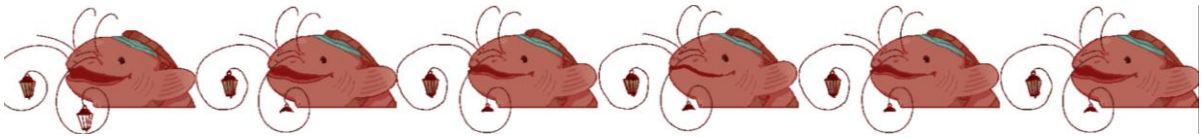
We decided on the animal the boss of each world represented at the very start of the year, for the boss of anger we settled on making it a caged dove, using the cage to protect itself and attack. The dove is white and gentle looking in its cage, the goal was to make it as soft and cute as possible. As it represents the ease and comfort one can find in anger when in the end It does nothing but destroy us from the inside.



One thing I added to show that the dove was hiding what it was is the sharp black spike on its back whenever it attacks, this is in reminiscence of the second enemy of anger who also had a black goop spike, showing the violence it hides beneath its soft exterior.

It is only when the dove is dead that it is revealed that the dove itself was also just a pile of black goop dissolving slowly into the ground back to the monster it came from. In this part we never really fight anger, this fight represents us accepting the fact that it exists within us. The dove was never really the final boss as it is just a part of the giant beast looming far away destroying the city.

3) Travelling Fish



The travelling fish was the first ever non-playable character I ever designed for this game. It is a catfish, friends of the animal of the forest and helps non fish creatures get around the forest using the river. In the game it is used as a checkpoint as well to travel to worlds the players have already completed.

The travelling fish is a giant catfish with lanterns at the end of its whiskers, as catfish can see very well in dark muddy water, the lantern is only for the passenger riding on top of it. I also put a saddle on top of it since it would imply him being a mode of transportation. Finally, for the animation on the sprite sheet I settled on having it move up and down in a rhythm.

4) Weapon Merchant

The weapon merchant does not actually sell weapons to the players but holds onto their equipment when not in use as the player can hold only one weapon at a time. It is a way to stock weapons. At first, we wanted it to be a giant frog that was quite shady looking and would sell the player weapon in a giant booth. In the end we figured having a mysterious figuring looming in the booth was at the same time easier and more interesting story wise. Which is exactly what representing the sprites. In the first sprite we can see the merchant at the booth in nothing but absolute darkness. When he is talking to the player his eyes light up.

C) Non-playable characters and Dialogues

I was tasked with implementing the non-playable characters in the game. I made an object NPC, where we can define different NPC's with their position on the screen as well as their sprite sheet and dialogue. In fact, implementing the non-playable characters meant I had to design the way these characters could talk. Non-playable characters had to also be packaged so that they could be sent as a dictionary on the server (more information on Pierre's parts in the rooms).

1) Handling the dialogue

The dialogues themselves will be handled by Eden and Tarek. But I had to figure out a way to implement it into the NPC object. Dialogues are inside a dictionary where each key is a conversation and item is a list of lists representing the different lines in the speech bubble for every bubble.

Whenever the player got close to the non-player character, when their hitboxes touched an interactive popup would appear on the screen on top of the character. When clicked on with the mouse the dialogue will start; every time the player clicks on the mouse a

new bubble appears on the screen. When there is no more text, the interact popup reappears and can be interacted with again by the player.

2) Cutscenes

Cutscenes are predefined scenarios, where the characters in it have a predetermined speaking pattern and the player can only skip ahead in the dialogue. To implement this, I reused the code for the room as if we were in the gameplay but instead of having two player objects, all the characters are NPC objects and therefore we can make them talk. At the start and end of every cutscene, there is a fade in and fade out transition to make it more visually appealing.

The first cutscene of the game happens right after the first flashbacks, when the Heart finds the girl on the shore of the lake, during this cutscene he explains what happened and that if they want to leave this forest, they need to follow the river down which creates the starting point of our game.

D) What needs to be done

The most important thing is to finish the game loop until the end of the game and through the end credits. So that we can have a fully playable game. I will work on making the game coherent between the parts of everyone in the group. I will also work on the many sprites that need to be done for the enemies and the different characters such as the oyster lady and the wandering merchant and the cards of the game. I also need to work on the savings of the game as for now we only have a basic saving system that saves the basic information such as the progression of the game. I also need to work on the randomization and the stat increase and effects of the cards. And finally have a basic and functional option page that we don’t have yet as we don’t have any modifiable parameters yet.

IV) Tarek's part

A) Graphics

1) Character design

Since the last defense, I was able to make progress on designing some characters for our video game, while finishing up on already established character ideas. Since our game is set on the riverside for the entirety of its play time, it was important to have character designs reminiscent of water spread across it. Thus came the idea of previously explored ideas such as the Nutria mob, whose designs I have finished and have a final character sheet for, which is featured below.

512 x 512

The nutrias now feature pieces of armor they wear as a sign of belonging to their group. They were made to look more robust, with their swords hanging in their back, as if they were barbarians. Although this character sheet features a nutria standing on all fours, their in-game design remains bipedal for the most part, as you will see further down in the sprites section.

As development of our game progressed, came the idea of implementing a "tarot card" mechanic inspired by the likes of ***Cult Of The Lamb*** and other roguelikes. As such, we came up with the idea of a new character to handle those cards: a clam lady. The idea was to use her pearl as her crystal ball, while the clam evokes a treasure chest, and the river is set near to.

Although the included image is a pretty rough sketch, we trust that the design is already quite well thought out and will look amazing once implemented in the game.

2) Sprites/Spritesheets

We have made a lot of progress regarding the assets we will use for our game, most notably on the enemy sprites and animations. So far, I have finished animating the nutrias' sprites, which include a basic walk cycle, as well as attack and death animations for them. Such animations are implemented through the use of spritesheets. They feature every frame making up the animation. Using pygame, we then cycle through those frames, and update the displayed image in order to create the illusion of motion.

Down below, the nutria's walk cycle in spritesheet format.

B) Music

Since the last defense, I have also started making music for our game. So far, I have finished the homescreen/menu music. In order to do so, I used Garageband on my iPhone. Most melodies were made through arranging different loops together, which heavily simplified the process. The drums were made by me, and the melody was later made and added onto them. The intention was to convey a calm, yet mysterious vibe, enticing enough to make the player want to dive into our carefully crafted world. The music will be handled by initializing the mixer module and triggering playback of sound files alongside the game loop.

V) Pierre’s part

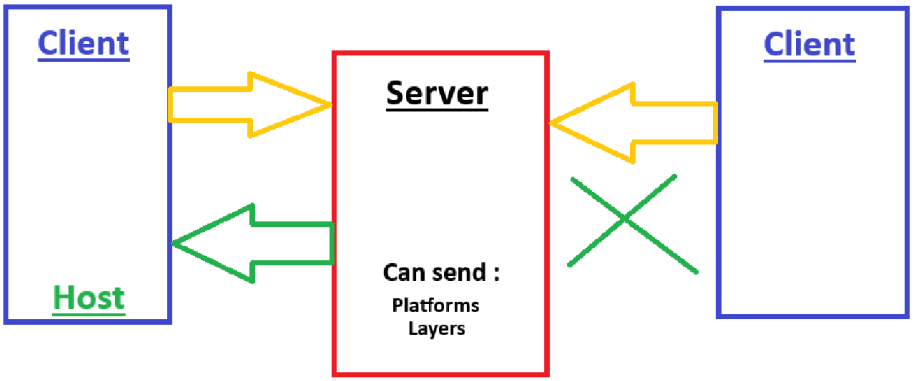
A) The room management system

While my teammates gave their all on the graphics of this game, I managed to improve the room management system I already created for the first part of the year. Before, the system worked like this : the players enter the game with both players able to see the same room (same platforms and graphisms) and the host of the game would press the SPACE button to change rooms, a new one being created from scratch taking randomly one image from each image folder related to the given world and assembling them.
This last part of our system still works the same: for the rooms where the player will fight small enemies, the rooms will be generated by taking one random background layer from all layer folders located in the associated world folder. That led us to a problem. How would we be able to create the same rooms over and over again if we wanted to have a lobby where players would return after each world completion or after their failure in a certain world? Because our old system didn’t handle pre-conceived rooms, I’ve had to implement a system that would allow our graphics team to fill different dictionaries and lists related to : the different layers they want based on the room they want to create (the merchants creek background for the lobby), but also the different npc’s in these predefined rooms. To finish this system, I also added a detailed commentary of this part of the code to make the process of filling the blanks easier than what it would be normally. I also added a way for the animated layers to actually play and not be static.

I’ve changed the way the transition worked: no need to press the SPACE key to change rooms anymore (this way was great for testing and debugging but highly impractical as a game feature and was never meant to stay forever). The players now must be together on the right side of the screen to initiate a transition and get to the next room. The difficult part has been to change the player’s positions from the right side of the screen to their initial spawn position (towards the left side of the screen) since the movement of the players I used to make the players go back to their original position didn’t match the one indicated in the player related aspect of the game. This is tricky but before: player positions used to be given by their position on an x axis, and their hitbox (a big rectangle fitting the player’s position) would be assigned to this x position. After some changes, the hitbox became the main position, and the x component adapted itself to the hitbox. I wasn’t aware of this and so spent some time figuring out what had happened until I finally understood and was able to complete this part. In addition to this right-transition feature, I’ve also implemented a way for the two players to go back to the previous lobby room so that they would be able to buy, discuss or interact in any way with the npc’s they might have missed.

B) Network

On paper, I had finished the server/client connection network system by the end of October 2025. In reality, the network is an issue on a day-to-day basis. The reason for this annoying part to be still a problem to consider today is that, in a multiplayer game, every action taken on one player or one player's screen has to be displayed on the second player's screen. And while this process is quite easy to understand, it is less easy to implement and handle when new systems and ideas are brought to life in the game. For example, after the creation of our first predefined rooms in the merchant's creek, only the first player was able to see the graphisms of the room, the second player not being able to see anything else than a black background with both players moving in it. To fix most problems, I created a Client_Events.py file and added some server events to the Server.py file to handle any movement from the players, any new room being loaded and generated and more generally speaking, any kind of change in the game will go through the network using events in the game. But this did not fix the problem with the predefined room not showing up on the second player's screen. To understand what went wrong, I'll have to dive a bit deeper in the actual code : when a new room is generated, the host will first ask for a new room to be generated by specifying if a room is a predefined room or a randomly generated room. The server will get this request and ask both clients to prepare for new incoming room data. The room data consisted at the time of only a list of positions of platforms and a list of all layers to load for the room. The problem was that for predefined rooms, the layers and the platforms are only a small part of the room data, the room name, id, the world the room is located in and the npc's configuration being a part of the data required to be sent for the clients to actually be able to load a new room. These data were sent to the host as it was the host that owned the server (and so had to receive the information he cared to receive). The other client, on another hand, never actually got to receive any data whatsoever, since there was no "hosting" link between them and the server. To resolve this problem, I've had to change the way the predefined rooms sent their data and the way the game and server handled these to actually make a link between the server and the second client.



This problem may seem very specific, but it is representative of most issues I've been fixing on the game

C) Music

While I haven't yet implemented my first audio for this game, I've created a soundtrack for the first world. This soundtrack will be an audio loop that plays continuously throughout the first world. The goal was to make a dark/somber but still enjoyable tune that should not bore or exhaust the player during the game. While this is easy to say, it is harder to make. Most of the time that I've spent on this task has been revolving around finding the best instrument that would fit these characteristics. The one I've chosen as of today is a glass harmonica, giving this echo-ey vibe that fits the somber/dark theme I went for (fitting for a world incarnating denial itself) if used in minor keys. The goal for this theme is to make it 1,30 minutes long to not create this feeling of boredom too rapidly for the players.
To compose this piece of the final soundtrack, I've used FL studio, one of the most popular music-making software programs in the world.

D) What needs to be done

I'll have to work on the item's implementation in the game now: hopefully the players will be able to pick up items after bosses die, buy items from the shop in the lobby, with some gold they will pick up after killing an enemy. I will have to work on the cards too to create a working system that feels fun to play and fair for the players.
I'll also have to create more music and probably increase the pace at which I'm making them (taking too long for a track isn't always a good thing, sometimes the music is already good and improving it won't change much). The color i wish to give these tunes will change based on the world the players are in (won't be the same music for anger than for sadness) but I will create a feeling of déjà-vu with some leitmotivs that will play throughout the whole game and vary slightly from song to song.
Finally, I'll stand ready and operational for my team if they ever encounter a network problem or any room related problem to fix these issues and bugs and give the best final product possible.

VI) Lucas's part

A) Enemies

Once the basis for the player was finished, came the time for the basis for enemies. We may have several different enemies planned, but they will all follow the same core principles, and I was tasked with making that core. Enemies move and act similarly to the players, although everything is predetermined by code instead of key inputs.

With enemies now in the game, the player needs to be able to fight them. For this, an attack zone was added and attached to the player's hitbox thanks to Pygame's useful Rect class. To attack, we make that attack zone "active" for a certain amount of time and delete the first enemy that comes in contact with it.

On the enemies' side, since they need a way to fight back, the same method was employed (attaching an attack zone to their hitbox). However, since we cannot press a button for the enemies to attack, they need a way to detect the players. This is done by checking if a player is in the zone, and if so, activate it. Although, if that were all, it would be highly unfair, as there would be no way to dodge. A cooldown was thus added, rendering enemies unable to attack sometime after an attack finishes, as well as adding a delay to their attacks, giving the players an opportunity to dodge, and potentially bait out attacks.

Making the enemies themselves was mostly straightforward with little hindrance, as they were like the player in many ways, as previously stated above. The bigger parts were the attack zones, detection and cooldowns. Getting everything just right wasn't easy and took a lot of trial and error.

B) Platforms

What took most of my time was implementing collisions with the various platforms in the room, added by other members of my group. Gravity and jumping themselves are complicated enough when having fixed ground, so the problem worsens when adding platforms, and many problems were encountered.

At first, there was a huge mismatch between coordinate systems: the players' coordinates were on an "unscaled" grid, while the platforms were in a "scaled" one. The scale here is to ensure screens of any size have the same visuals. This causes an impossibility to correctly compare the positions straight away. The fix here is straightforward but still requires an additional step of adapting one of the coordinates to the other format.

The second most prominent issue regarded the player not correctly stopping to move on contact with the platforms, and instead stopping the change in *gravity*, causing the player to phase through them at various speeds. Identifying the issue was a challenge, which was eventually solved.

Most other issues were mistakes on my part, such as inverting the adjustments to the player's position when they are above/under the platform, causing instant teleportation when touching the other side.

Ultimately, I had to rehaul how the movement works to more efficiently adjust the player's position. Before, the player's hitbox was adjusting to their position in coordinates (through a small calculation), and pressing movement keys adjusted the player's coordinates. Now, the player's hitbox is directly impacted by movement, and the player's coordinates are determined off the hitbox, effectively reversing the order of things.